

VISUALIZATION AND CHARACTERIZATION OF THE COVERAGE OF COMMERCIAL 4G BASE STATIONS

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1 Acknowledgment

I would like to express my special thanks of gratitude to my supervisor Mr. Xavier LA-GRANGE for providing me an opportunity to do this wonderful project on the topic 4G analyzing networks. He is also giving me all support and guidance which made me complete the project, although he had busy schedule managing the studies affairs. Secondly, I would also like to thank my colleagues and friends who helped me a lot in finalizing this project within the limited time frame.

2 Introduction

The fourth generation of mobile network technology, denoted as 4G or LTE, has a speed more than 10 times faster than its previous generation 3G. Nowadays, the popularity of 4G has increased in France, but the network quality as well as the cell plan is not announced by the operators.

In the terminal point of views, there are several cooperative applications that show the quality of 4G networks (opensignal, Nemo, ZKSAMp...etc.). However, theses application show the level of signal or the rate on a global level. The base station to which a mobile is connected is generally not reported and not shown. Our objective is to have a finer view of 4G networks: to show the different cells, to identify the radio site to which a terminal is connected to for a given location as well as the geolocation zone of the network

3 Description of tool and using database in the project

3.1 ZKSAMp

ZKSAMp is a compact, portable tool which collects RF network parameters by connecting to commercial phones (integrating QUALCOMM chipsets). Since this device connects to tablet, it allows collects the information in the communication between eNodeB and terminal, as well as the measurement through the radio connection. Then, it displays these results by the lines graph without PC.

ZKSAMp has its own GPS Antenna and five USB ports. The geolocation collected by ZKSAMp is more accurate than the geolocations from the others application using the GPS of mobile phone.

3.2 Cartoradio

<https://www.cartoradio.fr> is a website created by the National Frequency Agency. It lists all the antennas emitting more than 5 watts (except those of Civil Aviation, the Ministries of Defense and the Interior) in France. For this reason, we can find the information of all the operators's antennas such as geolocation,height, azimuth, system etc.

4 Data Preparation

Each site in Cartoradio is identified by an administrative reference. The measurement reports of ZKSAMp only includes the frequency number and a Physical Cell Identity (PCI). The same PCI value can be reused for different cells if they do not interfere with each other. Each cell is identified from a network point of view by a unique reference called Cell Identity (CID). There are no simple links between the administrative reference, the Physical cell Identity and the Cell Identity. The first step of the study is to associate a CID to a PCI and to an administrative reference.

4.1 CID - PCI association

Each eNodeB periodically broadcasts system information messages. These messages include a lot of parameters of the 4G network and of the cell: identity of the operator, configuration of the radio interface, signal level thresholds etc. In particular, the message called SystemInformationBlockType1 (see Figure 1) includes the CID, TAC of the cell. When a terminal is in a cell, it decodes the system information messages sent by the eNodeB. whenever the terminal access a cell, cell synchronization procedure is established. PCI value is encoded based on the primary and secondary synchronization signals. As ZKSAMp stores all messages that are decoded, we can find the CID. In addition, every report header, which is recuperated before converting this report in .pcap format, has a PCI value of cell serving. For that reason, we can make an association between PCI and CID.

Moreover, the ZKSAMp makes measurements as the reference signal received power (RSRP), the reference signal received quality (RSRQ) etc., on the serving cell every second. This measurement is associated to the PCI of the cell and stored in LTEPhone report (see Figure 2). Therefore, the combination of the LTEPhone and the SIB1 reports allows to generate a list of points in a cell. However, as it was mentioned before, two different cells can have the same PCI value if they have a sufficient distance, so figuring out the threshold value is necessary.

Suppose that a cell which its minimum area in the city centre is 1 km^2 and its shape is a hexagonal with the area is calculated by:

$$S = \frac{3\sqrt{3}r^2}{2} \approx 2.6r^2 \quad (1)$$

where r : radius of the hexagonal

The radius of the smallest cell is $\frac{1}{\sqrt{2.6}} \approx 620 \text{ m}^2$. Besides, in the EARFCN plan, there are only 504 PCIs, which lead to 504 cells. Thus, the minimum distance between two cells having the same PCI value is

$$r\sqrt{3}\sqrt{504} \approx 24\text{km}^2.$$

$\mathcal{S}_i$ can be defined as the convex envelopes of the measured cell i .

```

{
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  "_type": "pcap_file",
  "_score": null,
  "_source": {
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      "frame": {
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        "frame.interface_id_tree": {
          "frame.interface_name": "unknown"
        }
      },
      "frame.encap_type": "47",
      "frame.time": "Jul 31, 2018 14:41:22.000008000 Paris, Madrid (heure d\u00e2\u0080\u0099\u00c3\u00a9t\u00c3\u00a9)",
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      "frame.time_epoch": "1533040882.000008000",
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                      },
                      "item 2": {
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                      }
                    }
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}

```

Figure 1: Informations extracted from SIB1s

	PCI	EARFCN	Geolocation	RSRP
0	125	2825	POINT (48.107051 -1.665607)	51
1	125	2825	POINT (48.107049 -1.665608)	52
2	125	2825	POINT (48.107045 -1.665608)	47
3	125	2825	POINT (48.107043 -1.665607)	45
4	125	2825	POINT (48.107042 -1.665607)	49
5	125	2825	POINT (48.107041 -1.665611)	51

Figure 2: Information from LTEphone reports

	cellID	PCI	EARFCN	Geolocation	RSRP
0	('c0:fa', '01:06:60:10')	217	2825	POINT (48.115758 -1.691463)	-1
1	('c0:fa', '01:06:60:10')	217	2825	POINT (48.115758 -1.691463)	57
2	('c0:fa', '01:06:60:10')	217	2825	POINT (48.115758 -1.691463)	57
3	('c0:fa', '01:06:60:10')	217	2825	POINT (48.115802 -1.691425)	59
4	('c0:fa', '01:06:60:10')	217	2825	POINT (48.115814 -1.691412)	62
5	('c0:fa', '01:06:60:10')	217	2825	POINT (48.115837 -1.69139)	57

Figure 3: Information in a cell

- Suppose $\mathcal{S}_1$ and $\mathcal{S}_2$ have the same PCI value and $X_1 \in \mathcal{S}_1$ and P is the trace need to classify.
- if $d(X_1, P) < 24$, $P \in \mathcal{S}_1$, else $P \in \mathcal{S}_2$ where $d(X_1, P)$ is the distance between X_1 and P

The trace, which are measured from the SIB1, do not have the RSRP value, so -1 will be filled into the information cell table. (see Figure 3)

4.2 Approximate coverage of each base station

The website "https://www.cartoradio.fr" supports the information base station (in the CSV format) of the operators in France such as Orange, SFR etc in the Figure 4

In several countries, mobile operators transmit to a regulation entity the list of all their base stations with the geolocation of each one. The information is available in a public data base, which is called cartoradio in France. Thus, we assume that the geolocation $\mathbf{A}_i$ of each radio tower i is known.

The coverage of a radio tower can be determined by sophisticated planning tools that are feed by propagation models and digital elevation models. Our objective is not to

Numéro c	Numéro Cartoradio	Exploitant	Type d'ant	Numéro d'Ident	Dimension	Directivité	Azimuth	Hauteur /	Système	Début	Fin	Unité	Longitude	Latitude
30519	106454	SFR	Panneau	18618	2.7	Directif	120	28.4	LTE 800	801	811	MHz	-1.66278	48.1153
30519	106454	SFR	Panneau	18618	2.7	Directif	120	28.4	LTE 800	842	852	MHz	-1.66278	48.1153
30519	106454	SFR	Panneau	18618	2.7	Directif	120	28.4	LTE 1800	1730	1750	MHz	-1.66278	48.1153
30519	106454	SFR	Panneau	18618	2.7	Directif	120	28.4	LTE 1800	1825	1845	MHz	-1.66278	48.1153
30519	106454	SFR	Panneau	18618	2.7	Directif	120	28.4	LTE 2600	2500	2515	MHz	-1.66278	48.1153
30519	106454	SFR	Panneau	18618	2.7	Directif	120	28.4	LTE 2600	2620	2635	MHz	-1.66278	48.1153
30519	106454	SFR	Panneau	18619	2.7	Directif	0	22.4	LTE 800	801	811	MHz	-1.66278	48.1153

Figure 4: Informations of the sectors

predict in a precise way the coverage but to have a very simple method to determine a rough approximation of cells. We consider that each location is covered by the nearest radio site. In other words, the coverage of site i is the Voronoi cell related to $\mathbf{A}_i$ when considering all the sites of the area under study.

Due to the frequent use of sectorization, a radio site has generally 3 sectors and each one is associated with a single station from a functional point of view. In order to identify the coverage of a specific cell, the maximum and minimum azimuth of each cell site should be determined (see Figure 5).

Let L_i be the number of sectors on site i . Let $\beta_{i,j}$ be the azimuth of sector j for site i with $j \in [0, L_i - 1]$. Without loss of generality, we assume $\beta_{i,j} < \beta_{i,j+1}$ for $j \in [0, L_i - 1]$. Let $\alpha_{i,j}^{\min}$ and $\alpha_{i,j}^{\max}$ be the minimum and maximum azimuths of sector j in site i , respectively. For directional antennas, we have

$$\alpha_{i,j}^{\min} = \begin{cases} \beta_{i,j} - 60 & \text{if } L_i = 1 \\ \min(\frac{\beta_{i,j} + \beta_{i,j-1}}{2}, \frac{\beta_{i,j} + \beta_{i,j+1}}{2}) & \text{if } L_i > 1 \wedge j > 0 \\ \min(\frac{\beta_{i,j} + \beta_{i,L_i-1} - 360}{2}, \frac{\beta_{i,j} + \beta_{i,j+1}}{2}) & \text{if } L_i > 1 \wedge j = 0 \\ \alpha_{i,j-1}^{\max} & \text{if } L_i > 1 \wedge j = L_i - 1 \end{cases}$$

$$\alpha_{i,j}^{\max} = \begin{cases} \beta_{i,j} + 60 & \text{if } L_i = 1 \\ \max(\frac{\beta_{i,j} + \beta_{i,j-1}}{2}, \frac{\beta_{i,j} + \beta_{i,j+1}}{2}) & \text{if } L_i > 1 \wedge j > 0 \\ \min(\frac{\beta_{i,j} + \beta_{i,L_i-1} - 360}{2}, \frac{\beta_{i,j} + \beta_{i,j+1}}{2}) & \text{if } L_i > 1 \wedge j = 0 \\ \alpha_{i,0}^{\min} + 360 & \text{if } L_i > 1 \wedge j = L_i - 1 \end{cases}$$

In addition, for the isotropic antennas, we have:

$$\begin{cases} \alpha_{i,j}^{\min} &= 0 \\ \alpha_{i,j}^{\max} &= 360 \end{cases}$$

Numéro Cartoradio	Azimut	Système	↗	Longitude	Latitude	azimutLim1	azimutLim2
106454	120	LTE 1800		-1.66278	48.11528	60	180
106454	0	LTE 1800		-1.66278	48.11528	-60	60
106454	240	LTE 1800		-1.66278	48.11528	180	300
347610	0	LTE 1800		-1.65333	48.1175	-60	60
347610	120	LTE 1800		-1.65333	48.1175	60	180
347610	240	LTE 1800		-1.65333	48.1175	180	300

Figure 5: Azimuth limited of each cell

For each site, we can determine several angular sectors. Let $\mathcal{V}_{i,j}$ be the approximate cell related to cell i and sector j . Thus, $\mathcal{V}_{i,j}$ is the area defined by the intersection of Voronoi cell i and angular sector j . An example of approximate cells in Rennes city is given Figure 6. The green line represents the direction of the antennas and the triangle marker is the antennas associated with the measured cells .

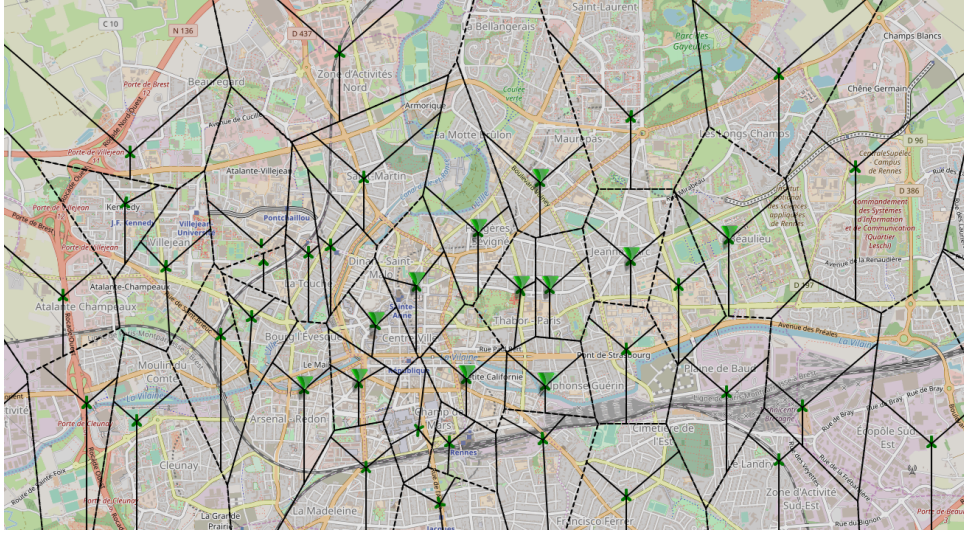


Figure 6: Voronoi diagram

5 Finding the relationship between cells and cells site

5.1 Set of measurement samples

With the field-test tools, we collect measurement samples. Each sample includes the identity of the cell the mobile is connected to, the measurement itself and the geolocation of the terminal. A *set* of measurement samples denoted by $\mathcal{S}_n$ is defined as the set all samples that include the same cell identity. To ease readability, we systematically use n to index a set of measurement samples and k to index a measurement sample.

5.2 Absolute Bearing calculation

Let $\mathbf{X}_k$ be the geolocation of the terminal for report k . Note that the geolocation of a point is the latitude and the longitude in radian denoted by ϕ_k and λ_k , respectively. Let $\mathbf{A}_i = (\phi_i, \lambda_i)$ be the geolocation of site i . Our objective is to find the sector of A that covers position X . We compute the *absolute bearing*, which is defined as the angle between the north direction and vector $\overrightarrow{AX}$.

Let $d_{N,i,k}$ and $d_{E,i,k}$ be the coordinates of $\mathbf{X}_k$ relatively to $\mathbf{A}_i$ in the North direction and in the East direction, respectively. They are given by

$$d_{N,i,k} = R(\phi_k - \phi_i), \quad (3a)$$

$$d_{E,i,k} = R(\lambda_k - \lambda_i) \cos(\phi_i) \quad (3b)$$

where R is the earth radius (see Figure 7).

Because the distance of the cell site to the measurement point is small compared to the earth radius, the absolute bearing $\alpha(\mathbf{A}_i, \mathbf{X}_k)$ from the cell size $\mathbf{A}_i$ to position $\mathbf{X}_k$ is in the flat earth approximation

$$\alpha(\mathbf{A}_i, \mathbf{X}_k) = \text{atan2}(d_{E,i,k}, d_{N,i,k}) \quad (4)$$

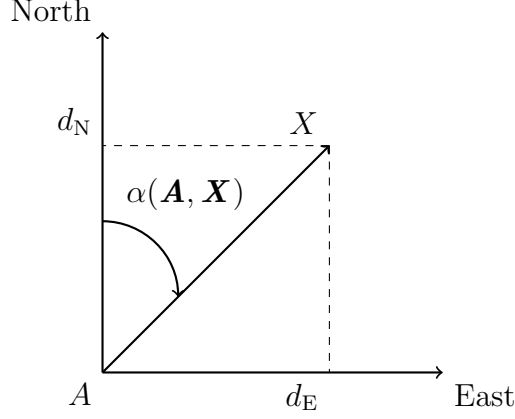


Figure 7: Bearing calculation

However, Since atan2 return values in the ranges $(-\pi; \pi]$, so the bearing from cell site to points in the first and second quadrant (correspond to the ranges $(0; -\pi)$) needs to be added 2π

5.3 Criteria calculation

The decision to attach a cell to a given identity cannot be taken on a single measurement sample but should take into account all the set.

Sample k is considered good if the RSRP value R_k is greater than -80 dBm and bad when it is smaller than - 100 dBm. Thus, we attribute a weight w of the k^{th} sample for the LTE signal is considered :

$$w_k = \begin{cases} 0.15 & \text{if } R_k \leq -100 \text{ dBm} \\ 0.9 & \text{if } -100 \text{ dBm} < R_k < -80 \text{ dBm} \\ 1 & \text{if } R_k \geq -80 \text{ dBm} \end{cases}$$

The first indicator is purely based on the angular sector defined by each base-station antenna

$$\sigma_{i,j,n} = \begin{cases} 0 & \text{if } \mathcal{V}_{i,j} \cap \mathcal{C}_n = \emptyset \\ \sum_{k \in \mathcal{S}_n} w_k \frac{\mathbb{1}_{\alpha_{i,j}^{\min} < \alpha(\mathbf{A}_i, \mathbf{X}_k) < \alpha_{i,j}^{\max}}}{\text{card}(\mathcal{S}_k)} & \text{otherwise.} \end{cases}$$

If there are a lot of samples that are in a given approximate cell, this means that this cell is a good candidate. The second criteria we consider is thus

$$\psi_{i,j,n} = \begin{cases} 0 & \text{if } \mathcal{V}_{i,j} \cap \mathcal{C}_n = \emptyset \\ \sum_{k \in \mathcal{S}_n} w_k \frac{\mathbb{1}_{k \in \mathcal{V}_{i,j}} \times \mathbb{1}_{\alpha_{i,j}^{\min} < \alpha(\mathbf{A}_i, \mathbf{X}_k) < \alpha_{i,j}^{\max}}}{\text{card}(\mathcal{S}_k)} & \text{otherwise.} \end{cases}$$

In order to get a simple decision criterium, the two indicators are merged into a global one:

$$\theta_{i,j,n} = \begin{cases} 0 & \text{if } \psi_{i,j,n} < 0.25 \\ 0.2\psi_{i,j,n} + 0.8\sigma_{i,j,n} & \text{otherwise.} \end{cases}$$

Where $\mathcal{C}_n$ is the interior of envelope convex $\mathcal{S}_n$.

In reality, there is always the overlapping among the close cells. As a result, the polygon convex from the measurement can be within several Voronoi cells. However, the overlap part is limited compared to the area of the cell, so the estimations which the ψ smaller than 0.25 are ignored.

5.4 Estimating the relation

Our plan firstly is finding the potential candidate sector in the view of measure cell and the good association will be selected in the view of the sector. The sector is chosen based on the threshold values θ_n of its association.

$$\begin{cases} \mathcal{A}_{i,j} \text{ is selected} & \text{if } \forall i, j, i', j', n < L, \theta_{i,j,n} \neq 0, |\log(\frac{\theta_{i',j',n}}{\theta_{i,j,n}})| > 0.1 \\ \text{Can't decide} & \text{Otherwise.} \end{cases}$$

where $\theta_{i,j,n} = \text{argmax}(\theta_{i',j',n})$ and L is the number of measured cells.

One sector could have various associations after the selecting step of cell measured selection. Thus, sector with various associations will be processed based on the minimum distance from site A_i to the envelope convex $\mathcal{S}_n$ ($d_{i,n}$) from site A_i

$$\mathcal{V}_{l,j} \text{ is selected if } d_{l,n} = \text{argmin}(d_{i,n})$$

6 Architecture of the software

6.1 Python

In python, there are 5 functions developed for converting .csv files to .Pcap files and preparing for the visualization in a web browser. In the following we describe each function and take an example to show the files that are manipulated or produced by the function. The field-test file produced by ZkSamp iszk_0000013797_20181031155939.txt

- Configuration: This function allows user to select the working directory which will store result files.
- Field-test*.txt to *.pcap and *.json conversion: This function allows user select a certain field-test file to preparing the .json field-test files and generating the .pcap file correspond to the operator(see Figure 9).The file created by ZkSamp includes the tests for all terminals (remember that ZKSamp is a multiterminal tool).The function generates as many .json files as the number of terminals for each field-test

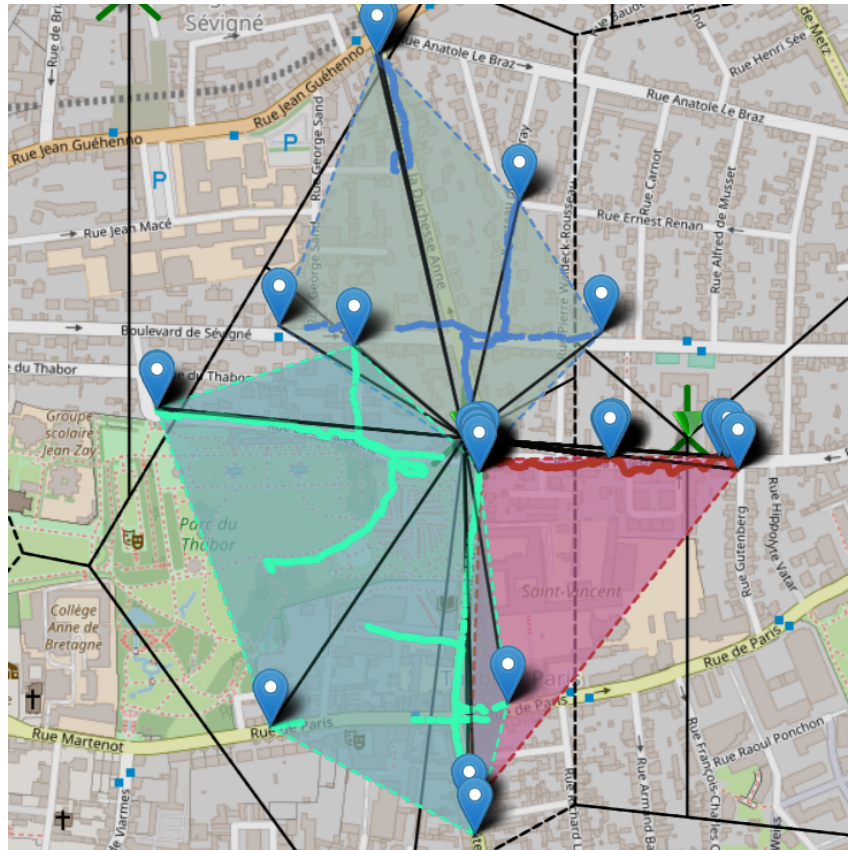


Figure 8: The cells and base station after estimating

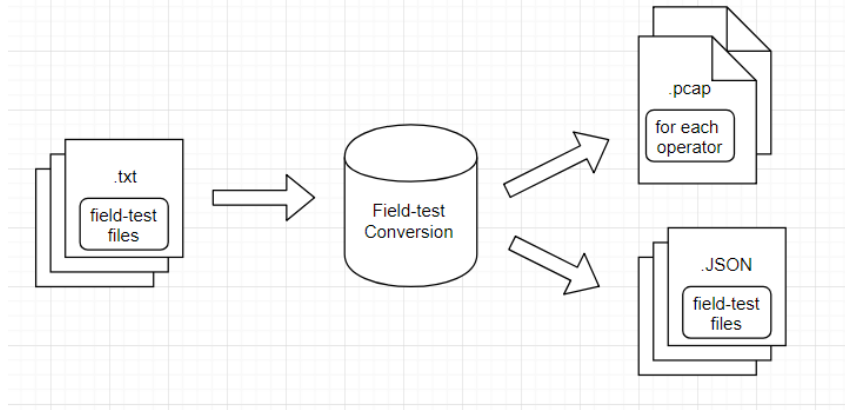


Figure 9: Diagram of Field-test Conversion Function

file. We assume the terminals have SIM cards from different operators. Hence, the files are renamed with the mobile country code and the mobile network code of the operator. This function merges all the .pcap files of the same operator into one .pcap file.

The diagram performed more details about this function is described in Figure 10. The selection protocol in the LTE Over-the-air message are discussed in the annexe. As we discussed before, each field-test file has multiple operators, so the first step of this function is classifying the operators. Then, for each operator, we filter the LTE Over-the-air message (whose the header type is "ML") and the LTE phone message (whose the header is "PL"). For the ML messages, the messages are classified and by different dissector based on transport channels and logical channels and written in a specific text format(discussed in the annexe). These text files are converted to .pcap files by application text2pcap and mergecap of WireShark. Then, the .pcap file are exported to .json ML file by using tshark. For the PL message, the messages are written in a .json format. The .json PL message and the .json ML file are used to create a .json field-test file which correspond to the fiel-test file from ZKSAMp.

For our reference example, this function generates the C208_01_zk_ 0000013797_ 20181031155939_ phone_1.json and C208_10_ zk _ 0000013797 _ 20181031155939 _ phone _2.json as the output with the field-test example.

- **Cartoradio File Conversion:** In cartoradio, there are multiple files, but the two files Supports-Cartoradio.csv and Antennes-Emetteurs-Bandes-Cartoradio.csv (remember that the second file contains all the information like azimuth, earfcn, system,...etc, but the geolocation of the sites is stored in the first file) used in this function for generating csv file and .json file using for Cell Association Processing step and visualisation respectively. (see Figure 11) Export the cartoradio files of the Calvados, the output of this function are sites_ORANGE.csv and sites_ORANGEE_Zone.json (for the Orange operator)
- **Cell Association Processing:** This function using the algorithm above for finding the relation between the sector and the cell measured and generating a .json file

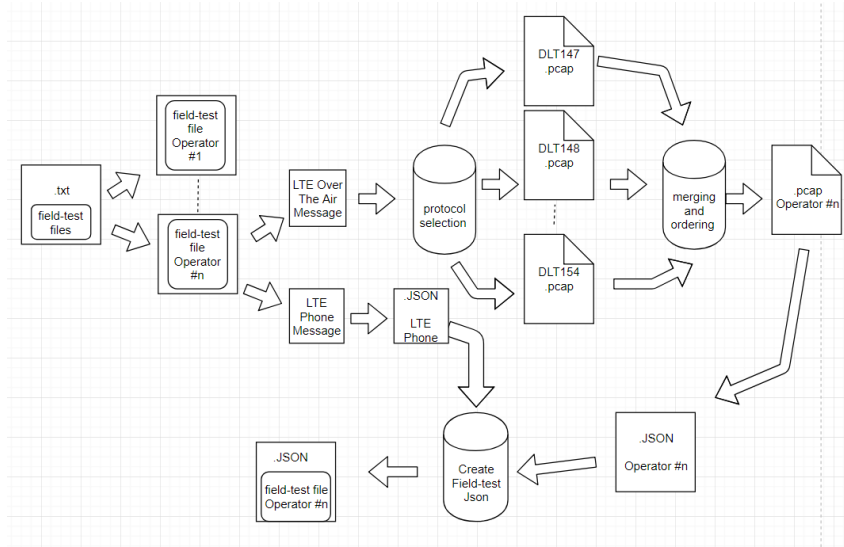


Figure 10: Diagram of Field-test Conversion Function(Detail)

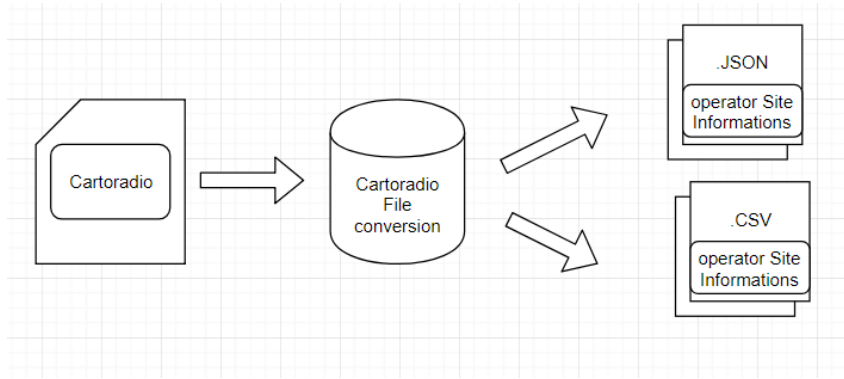


Figure 11: Cartoradio File Conversion

for visualization(.json format is mentioned in the Annexe). Thus, this function needs two files (the csv file from the Cartoradio file conversion output and the .json field-test files) for the input (see Figure 12).

Corresponding to the Orange operator, sites_ORANGE.csv and C208_01_zk_0000013797_20181031155939_phone_1.json are used as the input for this step, the output is ORANGE_association.json

- Visualization : This function allows user call html files from the Python interface.(see Figure 13)

6.2 JavaScript

This software using the openstreetmap for the visualisation. In the Cell characteristics table, some cell attributes are performed in the map like: PCI values, Tracking area regions serving cell RSRP and first neighbour RSRP offset.

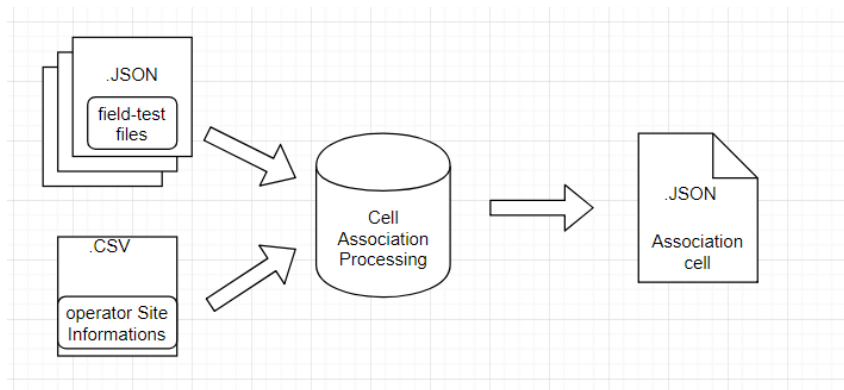


Figure 12: Cell Association Processing

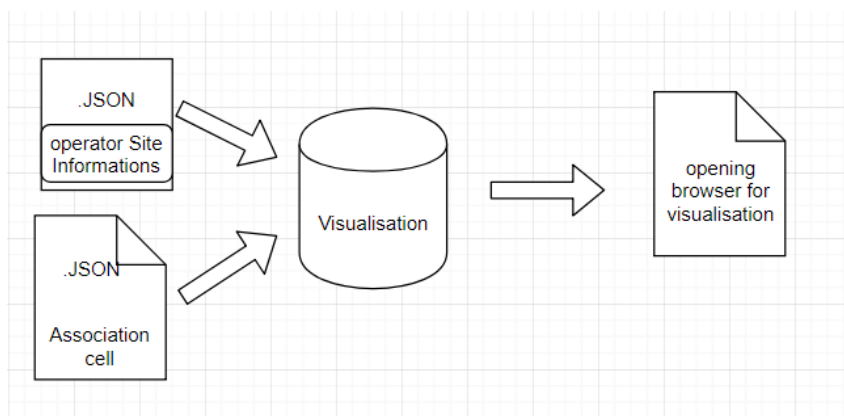


Figure 13: Visualisation cell coverage

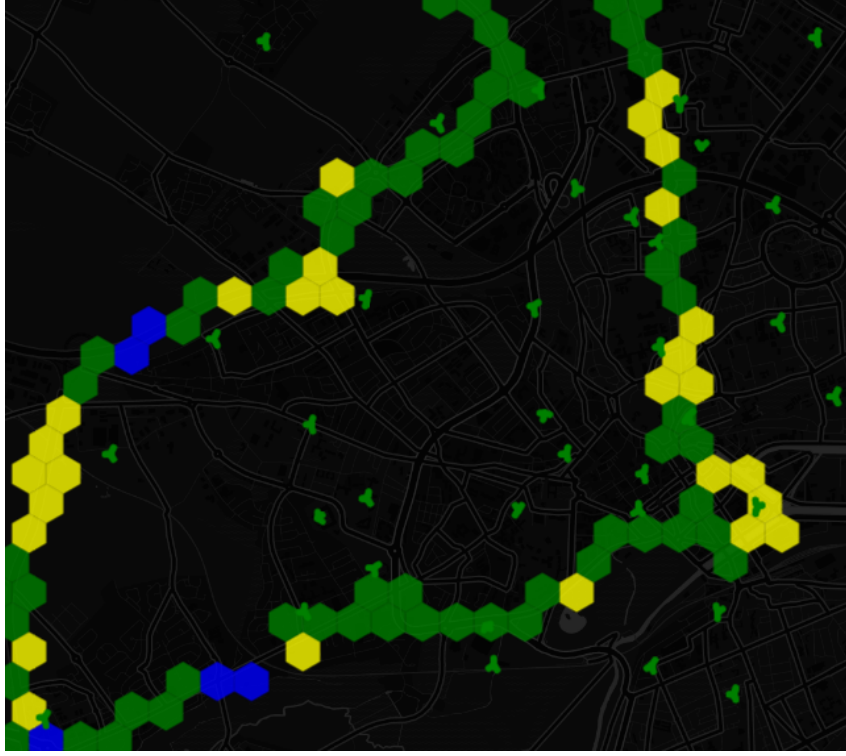


Figure 14: RSRP level of SFR operator

- Tracking Area Regions: In this function, the points measured is displayed with their tracking area (TA) value. The TA is encoded by 16 bits, but the color code need 24 bits. Moreover, two continuous tracking area do not have the big different in the TA values. Thus, I combined the first 8 bits with the last 2 bits of the second 8 bits to enhance their difference in the color. Suppose that the first eight bits of TA are $A_0, A_1..A_7$ and its second eight bits are $B_0, B_1..B_7$. The binary color code of the points measured will be : $A_0, A_1..A_7B_6, B_7000000B_7, B_6..B_0$
- PCI: In this function, the points in measured cell is filled with the PCI value. suppose that the PCI value has k bits, if PCI value is less than 256, the color code will be the combining of 2 bits in the left of PCI, (24-k) bits 0 and k bits of PCI. If PCI is greater than 255, the color code will be the combining of 2 bits in the left of PCI, 6 bits 0, 4 bits in the left of PCI, 4 bits 0 and the PCI value.
- Serving cell RSRP: This function displays the RSRP level in a region based on the minimum RSRP values in this region. (see Figure 14)
 - $\text{RSRP} < -121$:blue
 - $-121 \leq \text{RSRP} < -101$:green
 - $-101 \leq \text{RSRP} < -82$: yellow
 - $-82 \leq \text{RSRP} < -62$: orange
 - $\text{RSRP} \leq -62$: red



Figure 15: RSRP offset of SFR operator

- First neighbour RSRP offset: This function presents the difference of serving cell RSRP and the maximum of neighbor cell RSRP value in a region. Thus, we can compare the bandwidth between the different regions of an Operator (see Figure 15).

- RSRP offset < 0 :blue
- $0 \leq \text{RSRP offset} < 1$:cornflower blue
- $1 \leq \text{RSRP offset} < 2$: yellow
- $2 \leq \text{RSRP offset} < 3$: cyan
- $3 \leq \text{RSRP offset} < 4$:screamin green
- $4 \leq \text{RSRP offset} < 5$:green
- $10 \leq \text{RSRP offset} < 11$:yellow
- RSRP offset ≤ 17 :red

In the site table, the check box theory cell, we suppose that every sector has the same power and the same height, thus it displays the Voronoi diagram of the sites in the region with the direction of the sectors. If there are association cells, the serving site button allows user to choose polygon convex displaying with traces or not. If there are association cells, the landmark on the map allows user to display the cell in the EARFCN frequency corresponds to the tab EARFCN.

7 Conclusion

For this project, the visualization coverage of 4G Base stations was designed and programmed. Although it was designed with the log files of ZK-SAMp device, it has its

own input file structure. Test data were collected in Rennes city and Calvados city with Orange and SFR operators.

Comparing the results of the test data, in the cell association processing function, the SFR database is more accurate than the Orange database. Beside that, the tracking area of Orange Operator is smaller than the SFR's in the center of Rennes city.

The project can be continued by using the others measurement devices like Nemo, TEMS etc for collecting the data. So, some functions of the software should be separated for easier in the application upgrade. If using the project with mobile phone GPS, the geolocation needs to be corrected for more accuracy.

A Complementary information on formats

A.1 Text format for using text2pcap

The first line is the timestamps and the octaves in the second line are the messages. For example:

```
2018-03-27 06:14:13.000001
0000 68 48 20 03 38 A4 0D C5 20 39 1C 6A 98 48 40 C6 21 92 FD A0 00 00
2018-03-27 06:14:13.000007
0000 70 12 98 27 7D 92 11 83 81 BA 00 4C 00 48 00 C0 2F F9 4D 08 61 E1 3F 00 85
C2 83 87 00 E2 61 42 4C 11 27 96 BF 41 22 24
```

A.2 Protocol selection

I used WireShark application for convert the text file to .pcap file. So it is necessary to define the specific protocol by the option DLT_users in the Wireshark (see Figure 16)

DLT	Wireshark Dissectors
147	mac-lte-framed
148	lte-rrc.bcch.bch
149	lte-rrc.bcch.dl.sch
150	lte-rrc.pcch
151	lte-rrc.dl.ccch
152	lte-rrc.ul.ccch
153	lte-rrc.dl.dcch
154	lte-rrc.ul.dcch

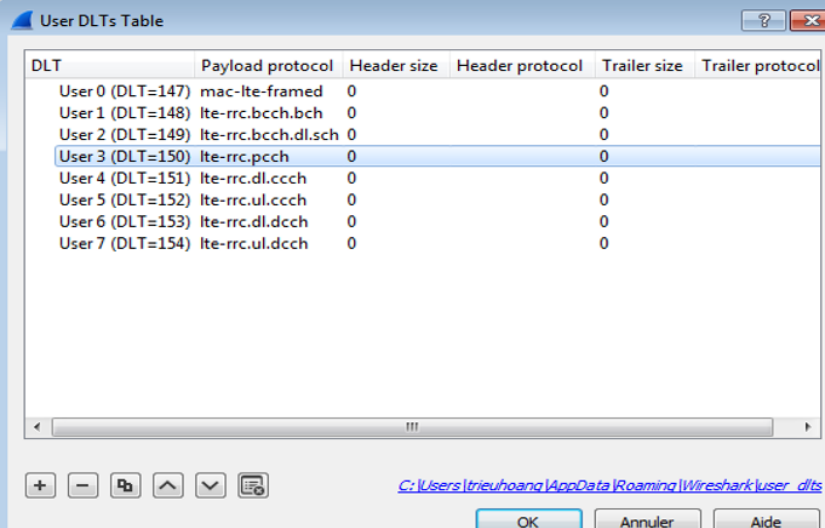
Figure 16: DLT_users dissector

From the logical channel field and the transport channel field in messages, the corresponding dissector of each message is selected (see Figure 17). In the Wireshark, the dissectors is configured like below:

- Edit -> preferences
- In the Preferences window, choose the DLT_User from the Protocols drop-down menu
- Choose the Edit option and set the parameters (see Figure 18)

Transport channels	Logic Channels					
	BCCH	PCCH	CCCH	DCCH	UL-SCH	MCCH
BCH	DLT(148)					
DL-SCH	DLT (149)	DLT(150)	DLT(151)	DLT(153)		
UL-SCH			DLT (152)	DLT (154)		
DL-MCCH						

Figure 17: dissector selection



DLT	Payload protocol	Header size	Header protocol	Trailer size	Trailer protocol
User 0 (DLT=147)	mac-lte-framed	0		0	
User 1 (DLT=148)	lte-rrc.bcch.bch	0		0	
User 2 (DLT=149)	lte-rrc.bcch.dl.sch	0		0	
User 3 (DLT=150)	lte-rrc.pcch	0		0	
User 4 (DLT=151)	lte-rrc.dl.ccch	0		0	
User 5 (DLT=152)	lte-rrc.ul.ccch	0		0	
User 6 (DLT=153)	lte-rrc.dl.dcch	0		0	
User 7 (DLT=154)	lte-rrc.ul.dcch	0		0	

C:\Users\trieuhoang\AppData\Roaming\Wireshark\user_dlt.s

+ - [Icon] [Icon] [Icon] [Icon] [Icon]

OK Annuler Aide

Figure 18: DLTs_users Table

A.3 .Json format of the Field-test file

In the "Mesurement" term, "RSRP" present the RSRP value in dB and "neighbourMax_RSRP" represent the maximum RSRP value of neighbour cells.

```
1  [
2      {
3          "SIB": {
4              "TAC": "30:05",
5              "CellID": "0e:72:90:10",
6              "PCI": "27",
7              "EARFCN": "1300",
8              "geolocation": {
9                  "lat": "49.316999",
10                 "lng": "-0.427664"
11             },
12             "mcc": "208",
13             "mnc": "01"
14         },
15     },
16     {
17         "SIB": {
18             "TAC": "30:05",
19             "CellID": "0e:72:90:10",
20             "PCI": "27",
21             "EARFCN": "1300",
22             "geolocation": {
23                 "lat": "49.316648",
24                 "lng": "-0.427871"
25             },
26             "mcc": "208",
27             "mnc": "01"
28         },
29     },
30     {
31         "Mesurement": {
32             "PCI": "53",
33             "EARFCN": "3000",
34             "Geolocation": {
35                 "lat": "49.178652",
36                 "lng": "-0.348474"
37             },
38             "RSRP": 53,
39             "neighbourMax_RSRP": 8
40         },
41     },
```

```

42     {
43         "Measurement": {
44             "PCI": "53",
45             "EARFCN": "3000",
46             "Geolocation": {
47                 "lat": "49.178636",
48                 "lng": "-0.348487"
49             },
50             "RSRP": 58,
51             "neighbourMax_RSRP": 17
52         }
53     }
54 ]

```

A.4 .Json format of the site file

suppose that the Box A bounds all the sites of cartoradio, and O is the point whose lat and lng are minimum. Divided A into 4 box equally (see Figure 19) the Node O is defined by latitude and longitude of point O, dlat and dlng correspond to the height and width of Box A. let the arrayPoint field be the stack which contains all of the geolocation of sites inside the zone A. The .json format of the site is defined below:

```

1 [
2     {
3         Node 0
4         "arrayPoint": [...]
5         "zone": [B,C,D,E]
6     }
7 ]

```

If the zone B has the dlat and dlng less than dlat_min and dlng_min respectively (I chose dlat is 0.5 and dlng is 3), the Zone of box B will be null ([]) The .json format association cells will be:

```

1 [
2     {
3         "lat": "48.833059999999996",
4         "lng": "-1.1047200000000001",
5         "dlat": "0.5808300000000006",
6         "dlng": "1.51333",
7         "arrayPoint": [...]
8         "zone": [{
9             "lat": "49.123475",
10            "lng": "-1.1047200000000001",
11            "dlat": "0.2904150000000003",
12            "dlng": "1.51333",

```

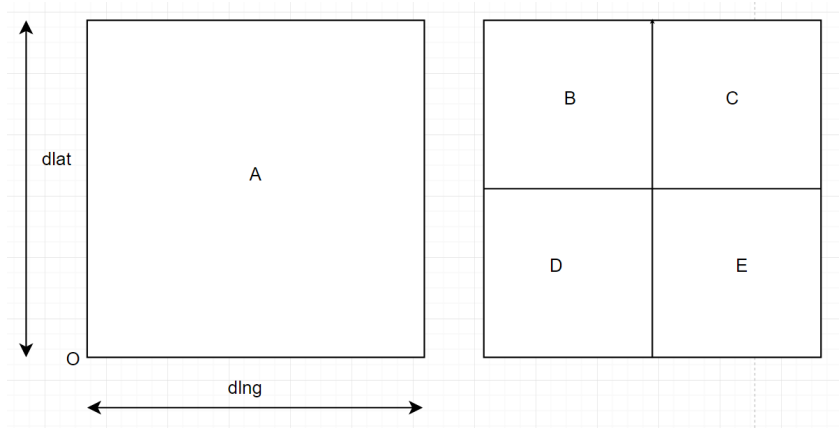


Figure 19: Zone inside .json format of the site file

```

13         "arrayPoint": [...]
14         "zone": [...]
15             }
16         ...
17     ]
18 }
19 ]

```

A.5 .Json format association cells

Every cell has 5 fields which are property, TAC, site, Azimuth, feature. The property contains 3 items which are CID, PCI and EARFCN. The site field presents Cartoradio number. The features field is a stack of measured points. In the measured point, "prop" field is the status of the point("inside" when this point is inside the polygon convex and "outside" when it is outside). The "RSRP" is the RSRP value(in dB) and the "RSRP_neighbour" is the RSRP offset (which is the difference of RSRP value of serving cell and the maximum RSRP value of neighbour cell). If The cell is not associated, the site and Azimuth field are filled by "None"

```

1 [
2     {
3         "property": "('84:60:e0:90', '391', '1501')",
4         "TAC": "79:58",
5         "site": 196159.0,
6         "Azimuth": "270.0"
7         "features": [
8             {
9                 "lat": 49.180074,
10                "lon": -0.349701,
11                "prop": "inside",

```

```

12         "RSRP": 48.0,
13         "RSRP_neighbour": 2
14     },
15     {
16         "lat": 49.180136,
17         "lon": -0.349745,
18         "prop": "inside",
19         "RSRP": 46.0,
20         "RSRP_neighbour": 2
21     },
22     ...
23 ]
24
25 ],
26
27 },
28 {
29     "property": "('84:60:e0:90', '391', '1501')",
30     "TAC": "79:58",
31     "site": "None",
32     "Azimuth": "None"
33     "features": [
34         {
35             "lat": 49.180074,
36             "lon": -0.349701,
37             "prop": "inside",
38             "RSRP": 48.0,
39             "RSRP_neighbour": 2
40         },
41         {
42             "lat": 49.180136,
43             "lon": -0.349745,
44             "prop": "inside",
45             "RSRP": 46.0,
46             "RSRP_neighbour": 2
47         },
48         ...
49     ]
50
51 ],
52
53 },
54 ...
55 ]

```